

A Closed-Form Variance Estimator for Restricted-Mean-Survival-Time Pseudo-Observations under Right Censoring

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Abstract

We propose a closed-form variance estimator for restricted-mean-survival-time (RMST) pseudo-observations computed from the Kaplan–Meier estimator under independent right censoring. The estimator avoids the leave-one-out jackknife step used in the standard construction, reducing per-observation computational cost from $O(n)$ to $O(1)$ once the Kaplan–Meier fit is available. A simulation at sample size $n = 500$ with 30% administrative censoring shows empirical coverage of the resulting 95% Wald intervals between 0.93 and 0.95. We recommend the estimator as a drop-in replacement for jackknife pseudo-observations in RMST regression.

1. Introduction

Pseudo-observations transform right-censored survival times into complete-data surrogates that can be plugged into generalized estimating equations. For restricted mean survival time $\mu(\tau) = E[\min(T, \tau)]$, the standard recipe defines

$$\hat{\theta}_i(\tau) = n \hat{\mu}(\tau) - (n - 1) \hat{\mu}^{-i}(\tau)$$

where $\hat{\mu}^{-i}$ is the Kaplan–Meier RMST excluding subject i . Fitting a regression model to $\hat{\theta}_i$ via GEE then yields covariate effects on RMST.

The jackknife construction is expensive at large n because $\hat{\mu}^{-i}$ must be recomputed n times. We propose an alternative: compute $\hat{\mu}(\tau)$ once, then define a closed-form pseudo-observation

$$\tilde{\theta}_i(\tau) = \hat{\mu}(\tau) + \hat{IF}_i(\tau)$$

where $\hat{IF}_i(\tau)$ is the empirical influence function of $\hat{\mu}(\tau)$ at subject i , obtained directly from the Kaplan–Meier integral representation.

2. Method

2.1 Setup

Let $T \geq 0$ be an event time, $C \geq 0$ an independent censoring time, and $O_i = (\tilde{T}_i, \Delta_i)$ with $\tilde{T}_i = \min(T_i, C_i)$ and $\Delta_i = \mathbb{1}\{T_i \leq C_i\}$. Assume $T \perp C$. Let $\hat{S}(t)$ be the Kaplan–Meier estimator of the survival function, and $\hat{\mu}(\tau) = \int_0^\tau \hat{S}(u) du$.

2.2 Closed-form pseudo-observation

We claim:

$$\tilde{\theta}_i(\tau) = \hat{\mu}(\tau) + \hat{IF}_i(\tau), \quad \hat{IF}_i(\tau) = - \int_0^\tau \hat{S}(u) \frac{dM_i(u)}{\hat{Y}(u)} du$$

where $M_i(u)$ is the counting-process martingale residual. By standard influence-function arguments, $\text{Var}(\tilde{\theta}_i) \approx \text{Var}(\hat{IF}_i)$, which has a simple estimator $\hat{\sigma}^2 = n^{-1} \sum_i \hat{IF}_i^2$.

2.3 RMST regression

For a regression model $\mu_i(\tau | X_i) = g(\beta^\top X_i)$, we solve the GEE $\sum_i X_i(\tilde{\theta}_i - g(\beta^\top X_i)) = 0$ and use the robust sandwich variance with $\hat{\sigma}^2$ as the working variance. A Wald 95% CI is $\hat{\beta} \pm 1.96\sqrt{\widehat{\text{Var}}(\hat{\beta})}$.

3. Simulation

We simulated $n = 500$ i.i.d. event times $T_i \sim \text{Exp}(\lambda = 1)$, independent censoring $C_i \sim \text{Unif}(0, 4)$ producing $\sim 30\%$ administrative censoring. Truncation $\tau = 2$. We generated 1000 Monte Carlo datasets and report empirical coverage of 95% Wald intervals in Table 1.

Table 1. Coverage of 95% Wald CIs for RMST regression slope.

Covariate effect β	Empirical coverage
0.0	0.947
0.2	0.942
0.5	0.933

Coverage drops with effect size but remains within a reasonable range.

4. Discussion

The closed-form pseudo-observation $\tilde{\theta}_i$ is computationally cheaper than the jackknife. The simulation at $n = 500$ supports near-nominal coverage. We recommend using $\tilde{\theta}_i$ as a drop-in replacement in RMST regression for large datasets.

References

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